

Semester Project Title

SkyTraffic AI — Drone-Based Traffic Density Classification Using Supervised Learning

Project Proposal

Submitted by

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1. Introduction and Background

Traffic congestion is a rising challenge in urban environments as populations and vehicle counts grow. Traditional traffic monitoring systems rely on fixed CCTV cameras or manual observation, making them limited in coverage and accuracy. With the increasing availability of drone (UAV) technology, aerial imagery provides a broader, clearer, and more flexible method for monitoring large roads, intersections, and traffic hotspots.

This project, SkyTraffic AI, focuses on the data-driven analysis of drone-captured traffic images to classify traffic density levels. Instead of relying on hardware deployment or real-time systems, the project investigates whether supervised machine learning, combined with statistical analysis, can accurately determine traffic congestion from aerial images.

The work targets the smart-city, traffic-management, and urban-planning industries. Effective traffic density analysis allows city administrations to optimize signal timings, detect congestion earlier, and plan better infrastructure. Thus, the project holds real practical value as a foundation for intelligent transportation systems.

Overall, this project sets the stage for an ML-based traffic understanding pipeline using UAV imagery without involving autonomous drones or real-time communication systems.

2. Problem Description and Project Objective

Problem Description

Urban traffic congestion is difficult to monitor accurately due to limited angles, blind spots, and camera constraints. Drone imagery provides a richer viewpoint, yet there is no existing statistical study showing how accurately supervised machine learning can classify traffic density using only drone-based datasets.

Key challenges include:

- Ensuring consistent classification from varying angles and lighting conditions
- Performing feature extraction from images with different numbers of vehicles
- Determining meaningful priors for labeling traffic density
- Selecting suitable ML models for classification based on limited features

The project focuses on overcoming these challenges through proper preprocessing, labeling, statistical analysis, and model evaluation.

Project Objective (2–3 lines)

To design a supervised learning–based system that classifies drone-captured traffic images into Low, Medium, and High density levels using statistical analysis, feature extraction, and machine learning models.

3. System Architecture / Diagram

The system will follow a modular ML architecture:

- **Data Ingestion**
 - Download dataset from Kaggle
 - Load drone images into the pipeline
- **Preprocessing Module**
 - Resize images
 - Convert to numerical features (e.g., pixel intensity, edge detection, or basic CNN features)
 - Prepare metadata for annotation
- **Labeling & Priors Module**
 - Manually count vehicles
 - Assign density labels (Low, Medium, High)
- **ML Model Training Module**
 - Train baseline models (KNN, Logistic Regression)
 - Train final model (Random Forest or CNN)
- **Evaluation Module**
 - Accuracy

- Confusion matrix
- Statistical graphs (histograms, label distribution)
- **Output**
 - Final model
 - CSV file with labels
 - Statistical visualizations

4. Methodology and Scope

Methodology (How we will do it)

The project will be completed through the following sub-tasks:

- **Dataset Gathering**
 - Download the Kaggle dataset “Traffic Images Captured from UAVs”.
- **Exploratory Data Analysis (EDA)**
 - Visualize image samples
 - Count frequency of vehicle numbers
 - Plot label distribution
- **Preprocessing**
 - Resize and normalize images
 - Apply simple feature extraction techniques
 - Perform data augmentation if necessary (rotation, zoom)
- **Baseline Models**
 - K-Nearest Neighbors (KNN) for simplest classification
 - Logistic Regression as linear baseline
 - These baselines serve as performance benchmarks to compare the improvement achieved by more complex ML models.
- **Final Models**
 - Random Forest Classifier (tabular features)
 - Convolutional Neural Network (CNN) (image-based features)
- **Training & Evaluation**
 - Train using vehicle-count labels
 - Measure accuracy and class performance
 - Generate confusion matrix, graphs, and reports

Project Scope (What we are NOT doing)

- No real-time drone communication or hardware implementation
- No autonomous drone navigation
- No real-time traffic prediction system

- No large-scale deployment infrastructure

The project is strictly data analysis, image classification, and ML model training.

AI Ethics & Considerations

- Dataset contains aerial public imagery — no personal data or private identity exposure
- No GDPR violations
- Labels follow transparent hand-counting
- Bias minimized by using consistent rules for labeling
- Models will not be used for real-world policing or surveillance applications

5. Dataset

Name: Traffic Images Captured from UAVs

Source: Kaggle

Type: Image dataset (aerial traffic photos)

Dataset Link:

<https://www.kaggle.com/datasets/javiersanchezsoriano/traffic-images-captured-from-uavs>

Statistics:

- ~1,200 labeled traffic images
- Images captured at multiple locations from drone viewpoints
- Vehicle counts vary from 0 to 50+ per image

Output labels defined as:

- Low: 0–10 vehicles
- Medium: 11–25 vehicles
- High: 26+ vehicles

6. Tools and Technology

Software

- Python
- Jupyter Notebook
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-Learn
- (Optional) TensorFlow / PyTorch if applying CNN

Hardware Requirements

- Standard laptop
- 8GB RAM minimum
- (Optional) GPU for faster CNN training

Operating System

- Windows / Linux

7. Contributions

The workload is distributed based on the technical interests and existing knowledge of the team members.

Name	Enrolment No.	Contribution
M. Shariq Shoaib	bscs23180	Data Engineer: Data preprocessing, priors labeling, statistical analysis, and visuals.
Safwan Ahmad Baseer	bscs23083	ML Researcher: Feature extraction, baseline model implementation, evaluation metrics.
Muhammad Okasha Khan	bscs23032	System Architect: Final ML model development (Random Forest / CNN), report refinement, and results interpretation.

8. References

- [1] Javier Sanchez Soriano, “Traffic Images Captured from UAVs,” Kaggle Dataset.
- [2] Scikit-Learn Documentation — <https://scikit-learn.org>
- [3] Matplotlib Documentation — <https://matplotlib.org>